

Liu Xinyao

Mobile: (+65) 8350 6100 | Email: liuz6710@gmail.com | Singapore PR | **Mandarin (Native) & English (Fluent)**

Technical Product Owner and B2B solution designer with 2 years of experience translating enterprise customer requirements into **technical specifications, integration designs and production rollouts**. Led discovery and delivery for **8 enterprise clients**. Hands-on with **SQL, Python and web technologies** for prototyping and troubleshooting. NTU EEE First Class Honours (Data Intelligence Processing).

WORK EXPERIENCE

PSA Corporation Limited, Singapore

Jul 2024 – Present

System Analyst (Product Owner)

► Customer Discovery & Technical Assessment

- **Led technical discovery for 8 enterprise clients**, including 4 requiring **custom integrations**; mapped customer workflows, existing systems and data constraints to define **solution scope and implementation requirements**.
- Sized a new e-commerce initiative at **S\$1.5M+ in projected revenue over five years** through merchant behaviour analysis and **presented the case to leadership**, winning prioritisation against competing projects.

► Technical Specification & Solution Design

- Translated customer requirements into **technical specifications** across 3+ internal and external systems over **REST/JSON APIs, EDI and SFTP** — selecting integration patterns (sync vs. async, polling vs. webhook), and defining request/response schemas, error codes, retry behaviour and validation rules for engineering to build against.
- Standardised pricing rules across 8 enterprise clients into a **reusable billing framework**, reducing manual invoice preparation by approximately **3 staff-hours per day per client** and **billing errors by approximately 20%**.
- Owned the **core data-model design** — a hierarchical structure tracking inventory below the customer-order level, so the platform could represent partial processing and returns while keeping inventory and charges accurate.

► Implementation Oversight & Delivery

- **Delivered the initial release of PSA's first Container Freight Station (CFS) digital platform in approximately two months**, leading specification reviews, UAT and rollout. Subsequently expanded the platform to support an enterprise logistics client handling approximately **1,500 containers per month**.
- Delivered post-launch **user training** and served as the ongoing **technical point of contact**, triaging issues and translating recurring customer feedback into platform improvements to support adoption.
- Own post-launch **system and data health**, using **advanced SQL** and logs to investigate anomalies. Traced a silent cross-system sync failure **orphaning ~100 records** to a non-atomic delete-then-create API pattern, then designed the compensating rollback logic that stopped it recurring.

Schaeffler Group, Singapore

Jan 2023 – May 2023

Analyst Intern — Mobile Robotics Team

- Built a **web-UI generation toolset** turning design mockups into reusable front-end components wired to backend services, enabling rapid prototyping of internal tools.
- Ran **user research** with robotics engineers and built a **real-time monitoring dashboard** (2D visualisation and 3D point-cloud rendering), improving system observability and debugging efficiency for mobile robot fleets.

PROJECTS

RoleScout — Job-Matching Proof of Concept (Python, SQL)

Personal Project, 2026

- Built a Python application that ingests ~4,000 job postings from **10 job-board and ATS APIs** into a **SQL database**, ranks them by semantic similarity and scores fit with an LLM — taking the idea from prototype to a working daily tool.
- Added **76 automated tests**, threshold-based evaluation checks and run-health monitoring; implemented retries with backoff and graceful degradation to handle API failures.

EDUCATION

Nanyang Technological University (NTU), Singapore

Aug 2020 – Jul 2024

Bachelor of Engineering (Electrical & Electronic Engineering), **First Class Honours**

CGPA: **4.50 / 5.00**

Specialization: Info-Communications Engineering (**Data Intelligence Processing**)

MOE Full Scholarship Recipient | Dean's List 2022–2023 | **Goldman Sachs Hackathon Top 4 SG** (2021)

SKILLS

- **Solution & Specification:** Technical discovery, requirements translation, technical spec & API contract writing, solution design for scalability & reuse, data modelling, UAT & go-live, stakeholder alignment across sales, operations and engineering
- **Technical:** **SQL** (window functions, CTEs, complex joins), **Python** (pandas, FastAPI, pytest), JavaScript / HTML / CSS (working knowledge), REST/JSON APIs & HTTP, EDI/SFTP, webhooks vs. polling, Git/GitHub, Power BI; proof-of-concept prototyping
- **AI:** LLM applications via API, semantic retrieval, prompt & context design, evaluation harnesses, human-in-the-loop calibration; Claude Code, Copilot